

Analysis of Key Drivers of Trading Performance

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Research Motivations

- Investigate liquidity demand and assess the degree of market participation: over-participation vs. under-participation
- Address several aspects reflecting trading performance:
 - Size of liquidity demand relative to predicted/actual traded volumes
 - Choice of execution strategy
 - Timing of the trade
- Aim: minimising impact on stock price and execution cost
 - Market impact is unanimously defined as a function of trading volume (or relative order size, i.e. the order size divided by the expected daily volume)
- Volume is a benchmark for determining optimal order size
- Volume (or liquidity) prediction plays a critical role in the investment process
- Very little research on the impact of calendar effects on the trading volume and investigating the European markets

Industry Background and Publications

- Industry background
 - Collaboration with Deutsche Bank
 - Expert guidance and validation of the industry necessity of this research
 - Providing solutions for practitioners to real-world problems
- Submitted publications
 - *Examining Drivers of Trading Volume in European Markets*
 - International Journal of Finance and Economics
 - *European Trading Volumes on Cross-Market Holidays*
 - Journal of Empirical Finance
 - *Expiry Day Effects on European Trading Volumes*
 - Empirical Economics
 - *Developing a Volume Forecasting Model*
 - International Journal of Forecasting
- Published (extra)
 - *Social media analytics: a survey of techniques, tools and platforms*
 - AI & Society, Springer

Research Objectives

- Investigate the key drivers affecting trading volume
- Further analyse notable calendar effects and their impact on trading volume
 - Day-of-the-week effect, the holiday effect, or the expiry day effect
 - Create an accurate non-trading calendar for the European exchanges
- Propose an out-of-sample model for volume prediction
- Benchmark the performance of several statistical methods and training window types

Research Methodology & Structure

- **Examining Drivers of Trading Volume in European Markets**
 - Exploration of market dynamics and drivers of trading volume
 - Stepwise regression and randomisation tests (overnight return intervening nights, and asymmetry)
- **European Trading Volumes on Cross-Market Holidays**
 - The effect of cross-market holidays on volumes
 - Ridge regression and randomisation tests (Cross-market holidays vs. regular Mondays, Monday bank holidays vs. weekend effect, and market capitalisation-based effect differentiation)
- **Expiry Day Effects on European Trading Volumes**
 - The effect of special events on the volumes: stock index futures expiries and MSCI quarterly rebalances
 - Stepwise regression and randomisation tests (Friday effect vs. futures expiries, and end-of-month effect vs. MSCI quarterly reviews)
- **Developing a Volume Forecasting Model**
 - Prediction accuracy for the size of a trade given the market context
 - Different training windows (moving vs. growing) of different lengths (1M, 3M, 6M, 1Y, 2Y)
 - Statistical methods:
 - OLS, stepwise regression, regularisation (L1 and L2), kNN (arithmetic mean and inverse distance weighting), SVR
 - Additional models: switching model (cross-stock), metamodel (stock-specific)

Experiment 1: *Examining Drivers of Trading Volume in European Markets*

Experiment

- In-depth exploration of potential drivers of trading volume
- Contribution of volatility and volume-price asymmetry
- Contribution of day-of-the-week effects

Models

- Volume model
- 'State A' model (volatility & volume-price asymmetry)
- 'State B' model (incl. overnight return)
- Day-of-the-week models
- Methodology:
 - Randomisation test for overnight return intervening nights
 - Stepwise regression
 - Randomisation test for asymmetry evaluation

Data

- 2,353 European stocks – daily market data.
- 1st January 2000 – 10th May 2015

Results

- The results identified some key volume predictors:
 - Lagged (autoregressive) volume time-series
 - Volatility features
 - More recent data
 - Day-of-the-week features
- Empirical evidence for the price-volume relation asymmetry (~70%)
- Day-of-the-week effects are stronger in conjunction with endogenous variables

Hypothesis	Proportion
Volatility improves the volume model	86.95%
More recent price information improves the model	87.21%
Historical dynamic model: intraday return asymmetry	56.53%
Historical dynamic model: overnight return asymmetry	54.07%
Historical dynamic model: total asymmetry	73.59%
Historical dynamic model better than raw day-of-week model	99.87%
Day-of-week features improve historical dynamic model	90.57%

Experiment 2: *European Trading Volumes on Cross-Market Holidays*

Experiment

- Cross-stock analysis, predicting the *relative* volume
- Cross-market holiday effect: lower volumes (-8.5%)

$$V_{rel} = \log \frac{V_{t0}}{\text{median}(V_{t-1}, V_{t-2}, \dots, V_{t-20})}$$

Data

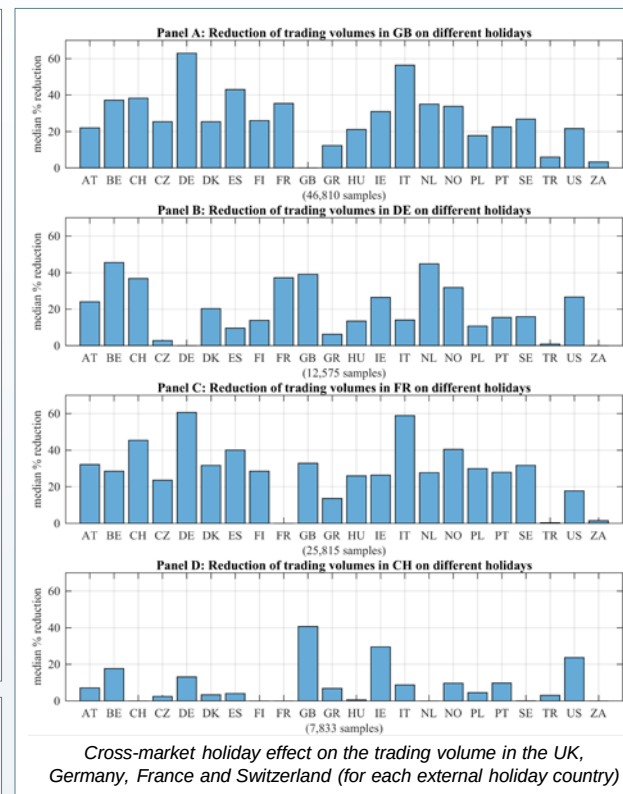
- Daily market data for 2,353 European stocks
- Accurate non-trading calendar for 21 European countries and USA (1st Jan 2000 – 10th May 2015)

Models

- Country-specific or pan-European (unified/global)
- Holiday country or holiday breakdown
- Multi-step ahead prediction
 - Performance decreases with the step size
- Methodology:
 - Randomisation test: cross-market holidays vs. control dates
 - Randomisation test: Monday bank holidays (mainly originating from the UK) vs. regular Mondays
 - Market capitalisation randomisation test
 - Ridge regression

Results

- Salient effects when the holiday takes place in a dominant market (USA, UK, Germany, Italy) or when most European markets are shut
- High susceptibility to cross-market holidays: Belgium, Spain, France, Hungary, Netherlands, Portugal
- The Monday bank holidays significantly drive lower volumes (compared to the weekend effect)



Experiment 3: *Expiry Day Effects on European Trading Volumes*

Experiment

- Anecdotal evidence of larger volumes on Fridays
- Cross-stock analysis, predicting relative volume

Models

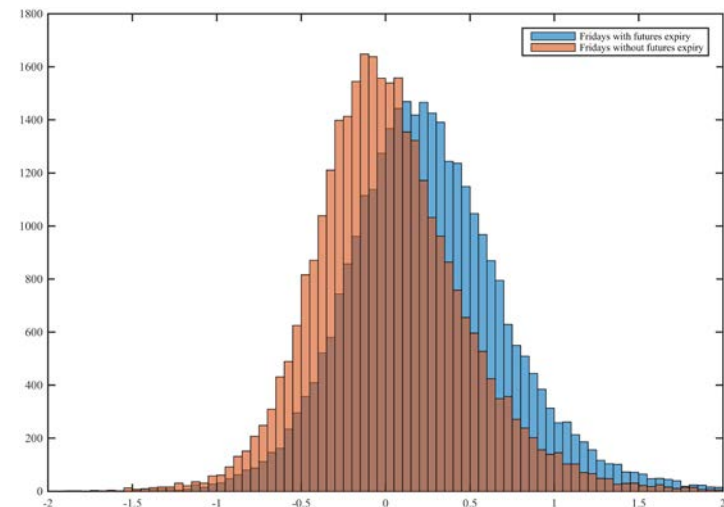
- Randomisation tests
 - Futures expiries vs. control dates
 - Friday effect vs. futures expiries
 - MSCI rebalances vs. control dates
 - End-of-month effect vs. MSCI quarterly reviews
- Stepwise regression:
 - Futures expiry models
 - MSCI rebalance models
- Multi-step ahead prediction, real-world use case

Data

- Accurate historical evidence of leavers and joiners for each index
- Futures expiries: 7 most liquid European indices (32,408 observations)
- MSCI rebalances: MSCI International Pan-Euro Index (10,298 observations)

Results

- Strong futures expiry effect, volume increases (-4d, +2d)
- Futures expiries significantly different from Friday effect
 - The expiries account for the larger volumes on Fridays
- MSCI has positive effect on volumes (-1d, rebalance day)
- MSCI rebalance months significantly different from adjacent months (with no rebalances)
 - MSCI rebalances not strong enough to explain the overall end-of-month effect throughout the year



Relative volume distribution for Fridays with futures expiries and Fridays with no futures expiries

Experiment 4: *Developing a Volume Forecasting Model*

Experiment

- Primary objective: optimal accuracy in predicting the size of a trade by improving the stability of error changes
- Propose a prediction model, trained on contextual data
- 33 years of runtime on HPC clusters

Models

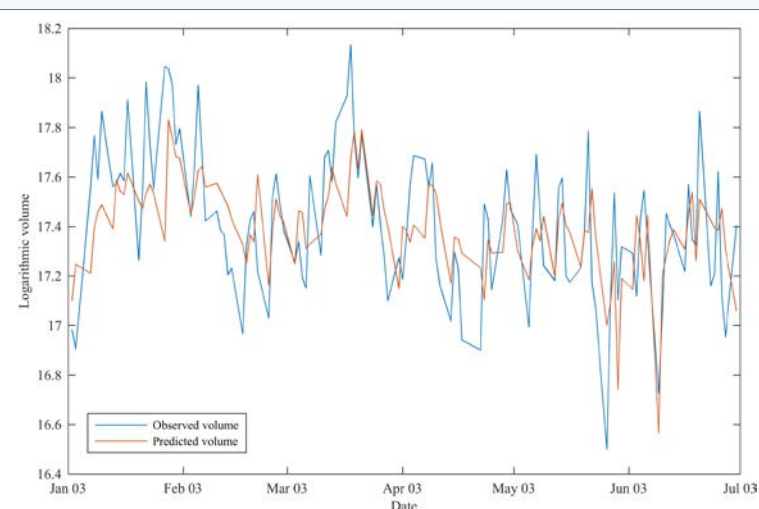
- With sliding/growing training windows of different sizes:
 - 1M, 3M, 6M, 1Y, 2Y moving vs. 2Y+ growing
- 42 stock-specific models for 7 statistical methods
 - OLS, stepwise regression, lasso, ridge, kNN with inverse distance weighting, kNN with arithmetic mean, SVR
- 9 cross-stock models for 3 event types:
 - Cross-market holidays, futures expiries, MSCI reviews
- Switching model
 - 42 event-oriented (cross-stock) submodels
 - Transition from one state to another based on the temporal context
- Metamodel
 - Stock-specific out-of-sample metamodel
 - Employs the stock-specific method that achieved best performance throughout recent time windows (1M/3M)

Data

- Calendar data (cross-market holidays, futures expiries, and MSCI quarterly reviews)
- OHLC prices and volumes for 2,353 European stocks, 1st Jan 2000 – 10th May 2015

Results

- Best performance: 2Y MW ridge regression
- Excellent performance by switching model



The volume prediction of the switching model for Telefonica SA

Contributions

1. Examining Drivers of Trading Volume in European Markets
 - Focus on the volume dimension
 - Detailed exploration of trading volume drivers
 - Comprehensive pan-European stock universe
2. European Trading Volumes on Cross-Market Holidays
 - Empirical evidence and quantification of the cross-market holidays
 - Accurate trading calendar data
 - Trading calendar for most liquid European exchanges and USA (22 countries)
3. Expiry Day Effects on European Trading Volumes
 - Empirical evidence and quantification of the expiry day effect
 - Accurate trading calendar data
 - European stock index futures expiries
 - MSCI quarterly review effective dates
4. Developing a Volume Forecasting Model
 - Out-of-sample volume model
 - Performance benchmarking for a variety of methods and training approaches
 - Established statistical methods applied to a new application domain